

Malcolm Greaves

MACHINE LEARNING SYSTEMS ENGINEER

U.S. Citizen, San Francisco Bay Area

☎ (414) 704-9696 | ✉ greaves.malcolm@gmail.com | 🌐 www.malcolmgreaves.io | 📱 malcolmgreaves | 📠 malcolm-greaves-49959919

Experience

Eventual, Inc.

San Francisco, CA

STAFF SOFTWARE ENGINEER

June 2025 - Present

- Built-out Eventual Cloud as the hosted Daft compute platform, securing a six-figure contract for enterprise deployment. Responsible for distributed runtime, egress, and control- and data- plane architecture for both hosted and BYOC environments.
- Contributed advanced PDF, image generation, common crawl dataset integration, & advanced type inference tutorials.
- Contributed distributed optimizations for unbalanced partition workloads to Daft, speeding up worst-case performance by 73%.

NVIDIA

New York City

SENIOR SYSTEMS ENGINEER IV

October 2023 - June 2025

- Created first automated, API-driven hardware and software stack testing service in NVIDIA Enterprise for BioNeMo. Ensuring that BioNeMo is compatible with dozens of NVIDIA GPUs, processor types, operating systems, versions of PyTorch and CUDA.
- Ongoing mentoring and skills-teaching to two dozen engineers and applied scientists. Weekly group demos and 1:1 teaching.
- Built-out BioNeMo's internal and external CI services to support open source development and internal testing and profiling. Led coordination efforts with internal infrastructure teams, optimizing scheduling and builds in CI, reducing build times by 30%.

Scale AI, Inc.

San Francisco & New York City

STAFF ML ENGINEER

March 2020 - October 2023

- Created Scale's first document processing models and supporting infra., securing a 7-figure contract for Scale Document.
- GPU util. cross-team initiatives: up leveled other ML eng to optimize their team's service from 20% to 86% GPU utilization.
- Advised and mentored ML engineer who optimized Kubernetes service start time from 2 minutes to under 30 seconds.
- Created ML model serving framework used by all ML engineers and hundreds of internal and external services.
- Optimized internal LLM services, achieving 10x speedup in end-to-end inference.
- Optimized real-time image generation + RAG pipeline with 10M+ requests/day with efficient vector DB: 50% cost reduction.
- Optimized batch embedding generation service used by multiple internal teams; decreased cost by 5x.

Change Healthcare

Emeryville, CA

SENIOR DATA SCIENTIST

November 2018 - February, 2020

- Developed deep NN model for predicting missing charges & deployed to US's largest medical insurance payment network.
- Increased scientist productivity by 25x with internal model training & serving framework.

Volley Labs, Inc.

San Francisco, CA

SENIOR RESEARCH ENGINEER, ML TECH LEAD

December 2016 - October 2018

- End-to-end ownership & development of multiple-choice question generation system from unsupervised to supervised neural network approach. Product-differentiating feature that helped Volley stand-out and make its first sale to JP Morgan Chase.
- Technical lead of engineering and ML team. Data pipelines in Airflow; deployed ML models using Keras, Spacy, Tensorflow.

Nitro Software, Inc.

San Francisco, CA

RESEARCH ENGINEER

March 2015 - October 2016

- Created novel machine learning based solution for automatic form field detection (FFD) and semantic classification. End-to-end service development & deployment to Nitro Cloud.

Alpine Data Labs

San Francisco, CA

SOFTWARE ENGINEER, MACHINE LEARNING

Jun. 2014 - Mar. 2015

- Algorithm optimization for large-scale data processing & modeling in Spark: L-BFGS, random forest models, feature encoding.

Education

Carnegie Mellon University

Pittsburgh, PA USA

B.Sc. AND M.Sc. IN COMPUTER SCIENCE

Aug. 2009 - May 2014

- Graduated with School of Computer Science Honors. GPA 3.5 (undergrad and graduate).
- Master's Thesis in semantic relation extraction from unstructured text with probabilistic logic & SVMs: <http://goo.gl/DzMr6c>

Work Portfolio

I am proficient across backend engineering, distributed systems, and machine learning (AI) engineering. This is the set of languages, tools, and technologies I am worked with professionally:

Languages	<i>Proficient:</i> Python, Rust, Scala, Go, SQL, Java, BASH, C; <i>Familiar:</i> Typescript, LaTeX, OCaml, C++11, CUDA
Machine Learning & Data	PyTorch, Pandas, HuggingFace, TensorFlow, torchvision, OpenCV, LangChain, torchaudio, numpy, scipy, scikit-learn, matplotlib, Onnx, Candle, burn, NVIDIA Triton, Deepspeed, Spark, AWS SageMaker, Apache Airflow, HuggingFace TGI, NVIDIA Dynamo, uv, llguidance
Infrastructure & Backend	Kubernetes, Datadog, Terraform, Istio, Temporal, Celery, Redis, Kafka, RabbitMQ, qdrant, milvus, AWS, CircleCI, Kubecost, Snowflake, Postgres, MongoDB, Protobuf, gRPC, OpenAPI, Flask, FastAPI, MinIO, DynamoDB, Postgres, MongoDB, Snowflake, Elasticache, MinIO, <i>Extensive Professional and personal experience with these Rust libraries:</i> tokio, rayon, serde, axum, crossbeam, datafusion, futures, reqwest, tower-http, tracing, rustls, kube, schemars, sqlx, chrono, anyhow, tungstenite